CS 6210: Matrix Computations

Review

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Big ticket items

Linear algebra and calculus

- Linear algebra background (abstract and concrete)
  - Vectors, spaces, subspaces, bases
  - Interpreting matrices: operators, mappings, quadratic forms
  - Canonical forms
- Calculus with matrices
  - Sensitivity analysis and conditioning
  - Variational notation for derivatives
  - Optimization with quadratics
  - Lagrange multipliers and constraints

Matrix algebra

- Ways to write matrix-matrix products
- Blocked matrices and blocked algorithms
- Graph structures: sparse, diagonal, triangular, Hessenberg, etc
- LA structures: symmetric, skew, orthogonal, etc
- Other structure: Toeplitz, Hankel, other special matrices

The big problems

$$\begin{aligned} Ax &= b \\ \text{minimize } &\|Ax - b\|^2 \\ Ax &= x\lambda \end{aligned}$$

The big factorizations

- LU and company (LDL^T and Cholesky)
- QR (economy and full)
- SVD (economy and full)
- Schur factorization
- Symmetric eigendecomposition

Iterations

- Iterative refinement
- Stationary iterations (Jacobi, Gauss-Seidel, etc)
- Krylov subspace definition
- Approximation from a subspace and Galerkin
- Characterization of CG and GMRES

Philosophical odds and ends

- Identifying the right structure matters a lot
- We need *both* algebra and analysis
- When you don't know what else to do... eigenvalues or SVD
- I differentiate five expressions before breakfast!

What else?

There is a lot that I wish I could get to in a course like this. If it were a two semester course, perhaps I would! Three things come immediately to mind.

- LA for data science (c.f. CS 6241)
  - Non-negative matrix factorizations
  - Tensors and tensor factorizations
  - More on factorization-based methods in stats/ML
  - The linear algebra of multivariate normals
  - Connections to convex optimization: active sets, quadratic programming, etc
- Iterative methods (c.f. CS 6220)
  - More on multigrid and domain decomposition
  - More on other "data-sparse" matrices
  - More on elliptic PDEs, integral equations, etc
- Eigensolvers
  - More on eigensolvers (especially iterative ones)
  - Much more on perturbation theory and sensitivity analysis
  - Matrix functions, and complex analysis connections
  - Connections to control theory
  - More on orthogonal polynomials

But there is always more to learn. If the course gave you a starting point to thinking about other corners of linear algebra that you care about for your research, then it was a success.

I enjoyed the class this semester. I hope you did as well.